

CHE 311 Homework Week 6

1. Using the continuity equation show that the flow defined by the following velocity field is possible.

$$\vec{v} = (2t + 2x + 2y) \vec{\delta}_x + (t - y - z) \vec{\delta}_y + (t + x - z) \vec{\delta}_z$$

2. A liquid ($\mu = 6.98 \times 10^{-3} \text{ lb-s/ft}^2$) is flowing at a rate of $0.2 \text{ ft}^3/\text{sec/ft.}$ of width of the plate between two flat parallel plates. Assume laminar flow. If the distance between the plates is 1 in., calculate:
 - 1) the maximum velocity
 - 2) the shear stress at the boundaries and
 - 3) the pressure drop in 10 ft. of length.
3. Consider the steady-state tangential laminar flow of a constant density and viscosity fluid between two vertical concentric cylinders. Assume that the inner cylinder is at rest. If the outer cylinder is rotating with an angular velocity ω , find:
 - 1) the velocity and shear stress distributions and
 - 2) the torque required to turn the outer shaft.
4. Two coaxial cylinders with radii R_0 and R_i rotate at angular velocities of ω_0 and ω_i respectively. Develop the velocity profile of the fluid filling the space between them. Assume isothermal laminar flow.